

SMU

System Management Unit

Beta release | Open-source RTL | Apache 2.0

HIGHLIGHTS

- Composes OCAH management, debug/test, and optional security subsystems
- Full reference configuration integrates SMC, SEP, DTP, and a 3-by-3 AXI4 fabric
- Alternate no-SEP configuration preserves SMC, DTP, and external AXI interfaces
- Runtime-programmed SMC/SEP apertures with fixed, loop-free connectivity
- Integrated reset, clock-stop, lifecycle, mailbox, interrupt, and debug handoffs
- Technology-neutral core plus an open reference wrapper for macro substitution

SYSTEM ROLE

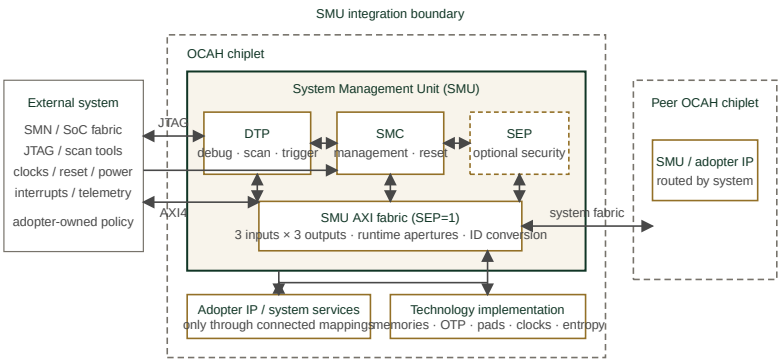
SMU is the composed management boundary between the chiplet system fabric, external debug and test equipment, adopter implementation resources, and the OCAH SMC, DTP, and optional SEP. It supplies their integration wiring and control handoffs; the adopter owns chiplet-level address assignment, clocks, power/reset sequencing, pads, technology macros, firmware policy, and product security authorization.

OVERVIEW

The System Management Unit (SMU) combines the System Management Controller (SMC), Debug and Test Ports (DTP), and, by default, the Security Processor (SEP). In the full configuration, a 3-input, 3-output AXI4 crossbar routes SMC, SEP, and external-system traffic using runtime-programmed SMC and SEP apertures. DTP connects JTAG, scan, reset override, cross-trigger, and debug-access paths; SMC coordinates management, resets, clocking, interrupts, and peripheral services; SEP supplies lifecycle-derived debug controls and security services.

The `smu` RTL top keeps technology-specific memories, OTP/eFuse, clocking, sensors, pads, entropy, and selected I/O behind typed integration interfaces. The open `smu_wrapper` attaches generic models and example termination for a reference build, but is not a characterized production implementation. Setting `SEP=0` removes SEP and the 3-by-3 crossbar, connects SMC directly to the stable external AXI boundary through ID converters, and applies defined SEP tie-offs.

SYSTEM CONTEXT



AT A GLANCE

Feature	Defaults
Headline configuration	<code>DefaultCfg, SEP=1</code> : 4-core SMC + SEP + DTP
Alternate configuration	<code>NoSepCfg, SEP=0</code> : 4-core SMC + DTP
Full-config fabric	3 AXI4 inputs by 3 outputs; 2 runtime address rules
AXI4 boundary	56-bit address, 64-bit data, 12-bit user
AXI IDs	8-bit input; 10-bit crossbar output; 6-bit SMC/SEP target
External interrupts to SMC	256
Mailbox instances	32 SMC; 8 SEP when present
External trigger interfaces	16 CTPs; 8 internal CTs; 8 clock-stop requests

`NoSepCfg` and `DefaultCfg` currently carry the same configuration fields; the separate `SEP` parameter selects presence. The adopter must record both values to identify a reproducible build.

CAPABILITIES

Composition and control

- SMC and DTP always instantiated; SEP selected at elaboration
- Internal wiring for DTP-to-SMC debug fabric and OTP, DTP-to-SEP debug and OTP, SEP-to-SMC mailbox/reset/security status, and SMC-to-DTP cross-trigger control
- SMC-owned primary reset sequencing, 32 subsystem reset/isolation slots, external boot/repair gate, watchdog handoffs, and DTP clock-stop output
- Lifecycle-derived SEP debug-disable controls feed DTP in the full configuration

Full-configuration fabric

- SEP can target SMC or the external output; SMC can target SEP or the external output; the external input can target SEP or SMC
- Runtime SMC and SEP aperture base/size rules; no same-port loops and no external-input-to-external-output path
- Unmatched SMC/SEP outbound traffic uses the external output; unmatched external inbound traffic receives a decode error
- AXI ID-width conversion at SMC/SEP targets; atomic operations disabled
- Fixed 1 GiB SEP-to-SMC local-alias window in the checked-in RTL

No-SEP behavior

- Direct external-to-SMC and SMC-to-external AXI paths retain the public widths
- SEP memory, entropy, I/O, mailbox, lifecycle-error, and status ports are tied to defined inactive values; the SEP aperture reports zero size
- SEP OTP debug accesses receive DECERR and the SEP secondary TAP has no target

INTERFACES AND CONFIGURATION

Feature	Integration options
Clocks and resets	SMU, reference, peripheral, telemetry, SEP watchdog, and entropy-sample clocks; cold reset, power-good, boot/repair, DFT, and reset outputs
System fabric	Inbound and outbound 56-address/64-data AXI4; programmable SMC and SEP aperture status; adopter AXI/AXI4-Lite extensions
Debug and triggers	Primary JTAG, boundary/IJTAG/secondary scan chains, external reset-control slice, 16 CTPs, 8 local trigger paths, and 8 clock-stop requests
Management	256 external interrupts, 32 reset/isolation slots, 32 SMC mailbox interrupts, GPIO/UART events, telemetry, timer, straps, and RAS status
Technology interfaces	SMC/SEP ROM and SRAM, eFuse/OTP, I3C table memory, trace memory, GPIO pads, PLL/PVT, entropy, and SEP crypto-memory boundaries
Build-time	SEP, Cfg, security-disable token, external entropy count, JTAG functions/identity, SMC CPU selection, and SEP memory/Adams Bridge options

INTEGRATION DEPENDENCIES

- Select `DefaultCfg/SEP=1` or `NoSepCfg/SEP=0`; preserve the complete parameter manifest with the release.
- Assign non-overlapping chiplet envelopes and program non-overlapping SMC/SEP apertures. Keep aperture controls stable while transactions are in flight and route unsupported atomic requests according to system policy.
- In `SEP=0`, DTP receives an all-zero debug-disable vector and the lifecycle output uses a fixed tie-off. Do not treat those values as product security authorization; define an acceptable no-SEP debug and lifecycle policy.
- Replace zero-valued JTAG identity and SEP security-token fields; supply reviewed firmware, provisioning, recovery, and update policy.
- Provide production clocks, reset/CDC constraints, power sequencing, pads, memories, OTP/eFuse, sensors, entropy, I/O, and physical implementation. Reference-wrapper models and error-slave terminations are examples only.
- Connect or terminate every external fabric, scan, trigger, interrupt, mailbox, reset/isolation, macro, and adopter-extension interface.

VERIFICATION AND MATURITY

The OCAH documentation website publishes the current verification and maturity status. Hosted cocotb/PyUVM smoke and regression use the bare `smu` DUT with `SEP=0`; the 53-test `sep0_all` group exercises SMC/DTP composition, direct external AXI, filters, ID conversion, atomic rejection, JTAG-to-AXI/OTP, reset, mailbox, cross-trigger, and clock-stop scenarios. This does not credit the full 3-by-3 fabric or live SEP interoperability.

A separate `smu_wrapper` catalog has four smoke entries spanning no-SEP and real-SEP profiles, including firmware and SEP boot-readiness observations. Additional real-SEP groups are separately enrolled or blocked and are not part of hosted SMU signoff. Wrapper results are not combined with bare-SMU results.

There is no agreed functional-coverage closure target or SMU RTL code-coverage configuration. No standards certification, silicon characterization, product security assessment, or portable frequency, area, power, latency, or throughput result is claimed.

DELIVERABLES

- `smu` RTL, packages, crossbar, and Bender manifests
- `smu_wrapper` and open IP-integration examples
- SystemRDL/generated registers and interface documentation
- Integrator guidance, specification, and synthesis constraints
- cocotb/PyUVM core/wrapper environments, firmware, and VPLANS
- Apache License 2.0 repository distribution

RESOURCES

Documentation: tenstorrent.github.io/tt-oca-harness
SMU specification: hw/sys/smu/doc/SMU_SPEC.md
Core / reference wrapper: hw/sys/smu/rtl/, hw/top/smu_wrapper.sv
Verification: hw/sys/smu/dv/
Integrator Guide: doc/integrator/
Project: github.com/tenstorrent/tt-oca-harness